t2-w09 — Simplification of Algebraic Terms — Simplification of algebraic expressions: order of operations, brackets and word problems

Run `c0c3-deepseek-native` · conditions C0, C1, C2, C3 · the pack shown is draw 3 of each.

HER LESSON PLAN

Simplification of Algebraic Terms — Simplification of algebraic expressions: order of operations, brackets and word problems

Objectives

1. carry out order of operation in algebra
2. insert/remove bracket and simplify expression
3. solve word problems involving algebraic expression

Step 0 — intro: activating prior knowledge

Teacher recalls with the students what they have already been taught, so the order-of-operations work has something to stand on:

Step 1 — order of operations in algebra

Teacher solves some questions in order of operation on the board while students listen, contribute, and where necessary ask questions where they do not understand, copy worked examples into their notes, and as well solve some questions independently.

- **teacher** solves some questions in order of operation on the board
- **student** listen, contribute, and where necessary ask questions where they do not understand
- **student** copy worked examples into their notes
- **student** solve some questions independently
- **teacher** moves round to monitor their work, answer their questions, mark their notes and mark correction

1. 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$

Step 2 — insertion and removal of brackets

Teacher solves some questions on insertion, removal of bracket on the board while students listen, ask questions on their areas of difficulty, copy worked examples into their notes and as well solve some questions independently. Teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections.

- **teacher** solves some questions on insertion, removal of bracket on the board
- **student** listen, ask questions on their areas of difficulty
- **student** copy worked examples into their notes
- **student** solve some questions independently
- **teacher** moves round to monitor their work, answer their questions, marks their notes and makes corrections

1. (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$

Step 3 — word problems involving algebraic expressions

Teacher solves some questions on word problems involving algebraic expression on the board while students listen, ask questions where they do not understand and as well solve some questions independently. Teacher moves round to monitor their work, answer their questions, mark their notes and mark correction.

- **teacher** solves some questions on word problems involving algebraic expression on the board
- **student** listen, ask questions where they do not understand
- **student** solve some questions independently

- **teacher** moves round to monitor their work, answer their questions, mark their notes and mark correction

1. 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.

Evaluation

1. Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ 2. Simplify $7 \times 3a - (3a + 5a) \times 2$ 3. Simplify $12 - (7 - 2f) - (3 - f)$ 4. Simplify $(8a - 5) + (14 - 3a)$ 5. A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils. **Summary.** Teacher does correction on the given exercises. **Conclusion.** Teacher marks students notes and give assignment.

1. Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$
2. Simplify $7 \times 3a - (3a + 5a) \times 2$
3. Simplify $12 - (7 - 2f) - (3 - f)$
4. Simplify $(8a - 5) + (14 - 3a)$
5. A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

HER LESSON NOTE

Verbatim, as transcribed. **C2 and C3 receive this; C0 and C1 never do** — which is the whole of what separates them.

Step 0 — intro: activating prior knowledge

```
1. Addition and subtraction of algebraic terms
2. Multiplication and division of algebraic terms
```

Step 1 — order of operations in algebra

```
#### Example 1. Find the value of the following

(a) $18 – 6 \times 2$

(b) $6 – 18 \div 3$

(c) $6 \times 2 – 2 – 40 \div 4$

**Solution**

(a) $18 – 6 \times 2$

$= 18 – 12$

$= 6$

(b) $6 – 18 \div 3$

$= 6 – 6$

$= 0$

(c) $(6 \times 2) – 2 – (40 \div 4)$

$= 12 – 2 – 10$

$= 10 – 10$

$= 0$

#### Example 2. Simplify the following

(a) $3x \times 2 + 5x$

(b) $17y – 5 \times 2y$

**Solution**

(a) $3x \times 2 + 5x$

$= 6x + 5x$

$= 11x$

(b) $17y – 5 \times 2y$
```

```
$= 17y - 10y$

$= 7y$

#### Example 3. Simplify the following

(a) $7x \div x + 5$

(b) $4x - (6 \div 3)$

(c) $(24x \div 6) + x + (1 \times 5x)$

**Solution**

(a) $7x \div x + 5$

$= \frac{7x}{x} + 5$

$= 7 + 5$

$= 12$

(b) $4x - (6 \div 3)$

$= 4x - 2$

(c) $(24x \div 6) + x + (1 \times 5x)$

$= 4x + x + 5x$

$= 5x + 5x$

$= 10x$

#### Activity. Simplify the following

1. $4n + 3n \times 10$
2. $32 + 8x \div 2$
3. $3 \times 5x + 4x \div 2$
4. $36y \div 6 + y + 2 \times 4y$
```

Step 2 — insertion and removal of brackets

```
**Hints**

- Always try to simplify the terms inside bracket first.
- A $+$ sign before a bracket does not change the signs of the terms inside the bracket, when
  bracket is removed.
- A $-$ sign before a bracket will change all the $+$ sign inside the bracket to $-$ and also
  change all the $-$ sign inside the bracket to $+$ when the bracket is removed.

#### Example 1. Simplify

(a) $a + (b + c)$

(b) $a + (b - c)$

(c) $2a - (3b - c)$
```

Solution

(a) $a + (b + c) = a + b + c$

(b) $a + (b - c) = a + b - c$

(c) $2a - (3b - c) = 2a - 3b + c$

Example 2. Simplify

(a) $a + (a - b)$

(b) $6x + (3 - x)$

(c) $(3a + 2) + (5a + a)$

Solution

(a) $a + (a - b)$

$= a + a - b$

$= 2a - b$

(b) $6x + (3 - x)$

$= 6x + 3 - x$

$= 6x - x + 3$

$= 5x + 3$

(c) $(3a + 2) + (5a + a)$

$= 3a + 2 + 5a + a$

$= 3a + 5a + a + 2$

$= 9a + 2$

Example 3. Simplify the following

(a) $5x - (3y + z)$

(b) $10d - (9c - 4d)$

(c) $(6x - y) - (4x - 5y)$

Solution

(a) $5x - (3y + z)$

$= 5x - 3y - z$

(b) $10d - (9c - 4d)$

$= 10d - 9c + 4d$

$= 10d + 4d - 9c$

```
$= 14d − 9c$

(c) $(6x − y) − (4x − 5y)$

$= 6x − y − 4x + 5y$

$= 6x − 4x + 5y − y$

$= 2x + 4y$

$= 2(x + 2y)$

#### Students activity

(a) $(12a − 7) + (4 − 5a)$

(b) $(8x − 2y) − (9x − 4y)$
```

Step 3 — word problems involving algebraic expressions

```
#### Example 1

A pen cost $P$ naira and a pencil cost 15 naira less than a pen.

(a) Find the cost of a pencil

(b) Find the cost of a pen and a pencil

**Solution**

Cost of a pen $= ₦P$

(a) cost of a pencil $= ₦(P − 15)$

(b) So cost of a pen and a pencil $= ₦P + ₦(P − 15)$

$= ₦P + ₦P − ₦15$

$= ₦2P − ₦15$

$= ₦(2P − 15)$

#### Example 2

A pencil cost $x$ naira and a rubber cost 40 naira more than a pencil.

(a) find the cost of a rubber

(b) find the cost of a pencil and a rubber

**Solution**

(a) cost of a pencil $= ₦x$

$\therefore$ cost of a rubber $= ₦(x + 40)$

(b) cost of a pencil and a rubber:

$= ₦x + ₦(x + 40)$
```

$= ₦(x + x + 40)$

$= ₦(2x + 40)$

Example 3

In a class, there are x boys and y girls last semester. This semester, 5 boys join and 3 girls leave the class. How many students are now in this class?

Solution

Total no $=$ number of boys $+$ number of girls

Number of boys $= x + 5$

Number of girls $= y - 3$

Total number $= x + 5 + y - 3$

$= x + y + 5 - 3$

$= x + y + 2$

Student activity

1. A pen cost h naira and a crayon cost ₦20 less than a pen.
   (a) Find the cost of a crayon
   (b) Find the cost of a crayon and a pen
2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes.
   Find the total cost of a pack of envelope and notepad.

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Simplification of Algebraic Terms

`subtopic`

Simplification of algebraic expressions: order of operations, brackets and word problems

`term`

Second Term

`week`

9

`lesson_objectives`

• carry out order of operation in algebra
• insert/remove bracket and simplify expression
• solve word problems involving algebraic expression

Objective 1 — carry out order of operation in algebra

`objective_text`

carry out order of operation in algebra

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — insert/remove bracket and simplify expression

`objective_text`

insert/remove bracket and simplify expression

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 3 — solve word problems involving algebraic expression

`objective_text`

> solve word problems involving algebraic expression

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

The Order of Operations in Algebra

EXPLANATION

When an algebraic expression contains several operations, we cannot just work from left to right. The result would depend on where you start, so everyone must follow one agreed order. You already know this order for numbers, and algebra follows exactly the same rules — only now the "numbers" may be letters that stand for numbers.

The order, often remembered as **BODMAS** (or **PEMDAS**), is:

1. **B**rackets — work out anything inside brackets first.
2. **O**rders — powers and roots, like \(x^2\) or \(\sqrt{x}\).
3. **D**ivision and **M**ultiplication — done together, left to right.
4. **A**ddition and **S**ubtraction — done together, left to right.

So in \(3 + 4x \times 2\), the multiplication is done before the addition: \(4x \times 2 = 8x\), then \(3 + 8x\). We never multiply by the 3 first, because addition comes last. In algebra, a term like \(4x\) is already a single amount (4 times \(x\)), so treat the letter as bound to its coefficient before adding or subtracting other terms.

EXPLANATION

Brackets tell you to do something first, but remember what they do: the number or term just outside a bracket multiplies everything inside it. Removing a bracket is called **expanding** it.

For example, \(2(x + 3)\) means \(2 \times x + 2 \times 3 = 2x + 6\). Multiply the 2 by every term inside the bracket.

When there is a minus sign in front of a bracket, be careful: it multiplies (or changes the sign of) every term inside. So \(-(x + 3) = -x - 3\), and \(-(2x - 5) = -2x + 5\). The minus turns a plus into a minus, and a minus into a plus, for each term.

After expanding, always look to combine any terms that are alike (the \(x\)-terms together and the plain-number terms together). This is how a tidy, single expression comes out at the end.

WORKED_EXAMPLE

Worked example 1 — operations in the right order

Simplify \(5 + 3x \times 4\).

- Multiplication comes before addition, so handle \(3x \times 4\) first: \(3x \times 4 = 12x\).
- Now add the 5: \(5 + 12x\).

The expression is already fully simplified. Note we do not write \(5 \times 3x\) and add a 4 — the order of operations prevents that. Final answer: \(12x + 5\).

WORKED_EXAMPLE

Worked example 2 — expanding and combining

Simplify \(3(2x + 4) + x\).

- Expand the bracket: \(3(2x + 4) = 3 \times 2x + 3 \times 4 = 6x + 12\).
- Bring down the \(+x\): \(6x + 12 + x\).
- Combine the \(x\)-terms: \(6x + x = 7x\).
- Final answer: \(7x + 12\).

WORKED_EXAMPLE

Worked example 3 — a minus in front of a bracket

Simplify \(4x - 2(x - 3)\).

- The \(-2\) multiplies everything inside the bracket. \(-2(x - 3) = -2x + 6\).
- Write the whole expression: \(4x - 2x + 6\).
- Combine the \(x\)-terms: \(4x - 2x = 2x\).
- Final answer: \(2x + 6\).

1. Simplify \(6 + 2x \times 5\).

answer: $10x + 6$

2. Simplify $2(3x + 5) + 4x$.
answer: $10x + 10$

3. Simplify $5x - 3(x - 2)$.
answer: $2x + 6$

4. Simplify $4 + 2(3x - 1) - x$.
answer: $5x + 2$

Brackets in Algebra: Opening and Building Them

EXPLANATION

When you see a bracket in an algebraic expression, it is a message: everything inside the bracket is one single term, and it is all being multiplied (or divided, or subtracted) by whatever sits just outside the bracket.

Think of a bracket like a box. The whole box is one object, no matter what is stacked inside it.

Removing a bracket (opening it)
A bracket with a plus sign in front comes off unchanged:
<code>3x + (2y + 5)</code> = <code>3x + 2y + 5</code>

A bracket with a minus sign in front comes off, and every sign inside flips:
<code>3x – (2y – 5)</code> = <code>3x – 2y + 5</code>

A bracket multiplied by a term outside it is expanded: the outside term multiplies each term inside, one at a time:
<code>2(x + 4)</code> = <code>2x + 8</code>

The sign of the outside term multiplies into the first term inside. So:
<code>–3(x – 2)</code> = <code>–3x + 6</code> because –3 × x = –3x and –3 × –2 = +6.

Putting a bracket in (building one)
Sometimes you go the other way and factorise: you find a term that divides every term in the expression, take it outside, and leave what is left inside the bracket.

To factorise <code>6x + 9</code>, look for the biggest factor of both 6 and 9 — that is 3. Every term divides by 3:
<code>6x + 9</code> = <code>3(2x + 3)</code>

To check: expand <code>3(2x + 3)</code> and you get back <code>6x + 9</code>.

To factorise <code>4x² + 8x</code>, every term has a factor <code>4x</code>:
<code>4x² + 8x</code> = <code>4x(x + 2)</code>

One rule to carry everywhere: what you take out of a bracket, you can always put back in. If expanding what you built does not return the original expression, the bracket is wrong.</p>

WORKED_EXAMPLE

Worked Example 1 — Remove the brackets and simplify
<code>5x + 2(3x – 4)</code>

Step 1: Expand the bracket. The 2 multiplies both terms inside:
<code>5x + 6x – 8</code>

Step 2: Collect the like terms (the <code>x</code>-terms together):
<code>5x + 6x = 11x</code>

Step 3: Write the answer with the number term after:
<code>11x – 8</code>

Worked Example 2 — A minus sign in front of a bracket
<code>8y – (3y + 2)</code>

Step 1: The minus sign flips every sign inside the bracket:
<code>8y – 3y – 2</code>

Step 2: Collect like terms:
<code>8y – 3y = 5y</code>

Answer: <code>5y – 2</code>

Worked Example 3 — Build the bracket (factorise)
Factorise <code>10x – 15</code>.

Step 1: The biggest factor of 10 and 15 is 5.
Step 2: Divide every term by 5: <code>10x ÷ 5 = 2x</code> and <code>15 ÷ 5 = 3</code>.
Step 3: Put the 5 outside the bracket: <code>5(2x – 3)</code>.

Check by expanding: <code>5(2x – 3) = 10x – 15</code>. ✓</p>

WORKED_EXAMPLE
Worked Example 4 — A word problem that builds a bracket
A rectangle has length <code>(x + 5)</code> cm and width 4 cm. Write the area of the rectangle as a simplified expression.

Step 1: Area of a rectangle = length × width:
<code>Area = 4(x + 5)</code>

Step 2: Expand the bracket:
<code>4(x + 5) = 4x + 20</code>

Answer: the area is <code>(4x + 20)</code> cm².

Worked Example 5 — Removing two brackets
<code>2(x + 3) – (x – 1)</code>

Step 1: Expand the first bracket: <code>2(x + 3) = 2x + 6</code>.
Step 2: The second bracket has a minus in front, so flip its signs: <code>–(x – 1) = –x + 1</code>.

Step 3: Put it together and collect like terms:
<code>2x + 6 – x + 1 = x + 7</code>

Answer: <code>x + 7</code>

1. Remove the brackets and simplify: <code>3(x + 6)</code>
answer: 3x + 18

2. Remove the brackets and simplify: <code>7y – (2y + 5)</code>
answer: 5y – 5

3. Build a bracket by factorising: <code>8x + 12</code>
answer: 4(2x + 3)

4. A rectangle has length <code>(3t + 4)</code> cm and width 5 cm. Write the area of the rectangle as a simplified expression.
answer: (15t + 20) cm²

5. Remove the brackets and simplify: <code>4(x – 2) + 3x</code>
answer: 7x – 8

From Words to Algebra: Solving Word Problems

EXPLANATION

A word problem describes a situation in words and asks you to find an unknown amount. To solve it, you first turn the words into an algebraic expression or equation, then simplify, and finally find the value you were asked for.

Here is a reliable set of steps to follow every time:

Step 1 – Read and decide what is unknown. Pick a letter for the amount you do not know (often x).

Step 2 – Turn the words into algebra. Break the sentence into pieces and write each piece using your letter. Watch for the words that mean operations:

"sum" and "more than" mean +
"difference" and "less than" mean −
"product" and "times" mean ×
"quotient" and "divided by" mean ÷

Step 3 – Simplify. Combine any like terms, and remove brackets carefully using the order of operations.

Step 4 – Answer the question. Check what the problem actually asked for, and state your answer with the correct unit (such as naira, metres or years).

EXPLANATION

Order of operations in algebra

When you simplify an expression with several operations, always follow this order:

Brackets first
Orders (powers and roots)
Division and Multiplication (left to right)
Addition and Subtraction (left to right)

This is often remembered by the word BODMAS. For example, simplify 2(x + 3) − 4. First do the bracket: 2(x + 3) = 2x + 6. Then subtract 4: 2x + 6 − 4 = 2x + 2. The bracket is removed, then like terms are collected, leaving 2x + 2.

WORKED_EXAMPLE

Worked example 1 – Finding an unknown number

The sum of a number and 5 is 17. Find the number.

Step 1: Let the unknown number be x.
Step 2: "The sum of a number and 5" means x + 5. "Is 17" tells us this equals 17:
x + 5 = 17
Step 3: Subtract 5 from both sides:
x = 17 − 5 = 12
Step 4: The number is 12.

WORKED_EXAMPLE

Worked example 2 – Two steps with a bracket

Twice a number is increased by 3, giving a result of 15. Find the number.

Step 1: Let the number be x.
Step 2: "Twice a number" is 2x. "Increased by 3" adds 3, and the result is 15:
2x + 3 = 15
Step 3: Subtract 3 from both sides:
2x = 15 − 3 = 12

Now divide both sides by 2:
x = 12 ÷ 2 = 6
Step 4: The number is 6.

WORKED_EXAMPLE
Worked example 3 – Removing a bracket in a word problem

A shopkeeper sells pens. On Monday she sells x pens. On Tuesday she sells 4 more than twice Monday's amount. Write an expression for Tuesday's pens and simplify it.

Step 1: Monday's pens = x.
Step 2: "4 more than twice Monday's amount" means 2x + 4.
This expression is already a single term sum; there is no bracket to remove and no like terms to combine.
Answer: Tuesday's pens are 2x + 4.

Now suppose the shopkeeper sells pens for 5 naira each. The money she earns on Tuesday is 5 times the number of pens, written with a bracket:
5(2x + 4)
Remove the bracket (this is the "remove bracket" skill):
5(2x + 4) = 5(2x) + 5(4) = 10x + 20
So Tuesday's earnings are 10x + 20 naira.

WORKED_EXAMPLE
Worked example 4 – A word problem you solve fully

A rectangle has length 3 metres longer than twice its width. If the width is w metres, write the length, then find the length when the width is 4 metres.

Step 1: Width = w.
Step 2: "Length is 3 metres longer than twice the width": 2w + 3.
Step 3: There is no bracket to remove. To find the length when the width is 4, substitute w = 4:
2(4) + 3 = 8 + 3 = 11
Step 4: The length is 11 metres.

1. The product of a number and 4 is 36. Write this as an equation and find the number.
answer: Let the number be x. "Product of a number and 4" means 4x, which equals 36, so 4x = 36. Dividing both sides by 4 gives x = 9. The number is 9.

2. A number is doubled and then decreased by 7 to give 11. Write the equation and find the number.
answer: Let the number be x. Doubled means 2x, decreased by 7 means –7, giving 11: 2x – 7 = 11. Add 7 to both sides: 2x = 18. Divide by 2: x = 9. The number is 9.

3. A pencil costs x naira. A pen costs 5 naira more than twice a pencil. Write an expression for the cost of one pen, then simplify 3 times that expression (the cost of 3 pens) by removing the bracket.
answer: One pen costs 2x + 5 naira. Three pens cost 3(2x + 5) = 6x + 15 naira.

4. A boy is y years old. His father is 20 years older than three times the boy's age. Write an expression for the father's age, then find the father's age when the boy is 6 years old.
answer: Father's age = 3y + 20. When y = 6: 3(6) + 20 = 18 + 20 = 38. The father is 38 years old.

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Simplification of Algebraic Terms

`subtopic`

Simplification of algebraic expressions: order of operations, brackets and word problems

`term`

Second Term

`week`

9

`lesson_objectives`

• carry out order of operation in algebra
• insert/remove bracket and simplify expression
• solve word problems involving algebraic expression

`entry_behaviour`

• Addition and subtraction of algebraic terms
• Multiplication and division of algebraic terms

`instructional_materials`

• Chart showing algebraic expression.

Objective 1 — carry out order of operation in algebra

`objective_text`

carry out order of operation in algebra

`current_step`

1

`instructional_events`

- provide_learning_guidance
- elicit_performance
- provide_feedback

`student_tasks`

- 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$

`assessment_points`

- Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$
- Simplify $7 \times 3a - (3a + 5a) \times 2$

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — insert/remove bracket and simplify expression

`objective_text`

insert/remove bracket and simplify expression

`current_step`

2

`instructional_events`

- provide_learning_guidance
- elicit_performance
- provide_feedback

`student_tasks`

- (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$

`assessment_points`

- Simplify $12 - (7 - 2f) - (3 - f)$
- Simplify $(8a - 5) + (14 - 3a)$

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 3 — solve word problems involving algebraic expression

`objective_text`

solve word problems involving algebraic expression

`current_step`

3

`instructional_events`

• provide_learning_guidance
• elicit_performance
• provide_feedback

`student_tasks`

• 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.

`assessment_points`

• A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Which Operation Comes First in Algebra?

EXPLANATION

When an algebraic expression contains more than one operation, you cannot just work from left to right. You must follow a fixed order, the same order you already use with numbers. That order is:

Brackets first, then of (multiplication in a bracket), then division and multiplication (working left to right), then addition and subtraction (working left to right).

A handy way to remember it is BODMAS: Brackets, Of, Division, Multiplication, Addition, Subtraction.

Division and multiplication are done together, from left to right, whichever comes first. Addition and subtraction are also done together, from left to right. The letters a and s at the end do not mean addition always beats subtraction, any more than the letters d and m mean division always beats multiplication. Left to right decides between the two in each pair.

The letters in the expression stand for numbers, but the rules do not change: you treat 3n exactly as you would treat a number times 3.

EXPLANATION

Let us see how BODMAS settles an expression like 4n + 3n \(\times\) 10.

There are no brackets, so the bracket step is skipped. Multiplication comes before addition, so the first thing to do is 3n \(\times\) 10. Since 3n means 3 \(\times\) n, this is 3 \(\times\) n \(\times\) 10 = 30n.

Now only the addition is left:

4n + 30n = 34n.

So 4n + 3n \(\times\) 10 = 34n. Notice what changes if you ignore the order: doing 4n + 3n first would give 7n, and then 7n \(\times\) 10 = 70n — a completely different (and wrong) answer. Order matters.

WORKED_EXAMPLE

Worked example. Simplify 32 + 8x \(\div\) 2.

No brackets here, so division comes first. 8x \(\div\) 2: since 8x means 8 \(\times\) x, this is 8 \(\div\) 2 times x, which gives 4x.

Now add: 32 + 4x = 32 + 4x. Because 32 is a plain number and 4x is a term in x, they cannot be combined. The answer is 32 + 4x.

WORKED_EXAMPLE

Worked example. Simplify 3 \(\times\) 5x + 4x \(\div\) 2.

Multiplication and division both appear, and neither is in a bracket. Do them together, left to right.

First, from the left, 3 \(\times\) 5x = 15x. Then 4x \(\div\) 2 = 2x.

Now the expression reads 15x + 2x, and only addition is left:

15x + 2x = 17x.

So 3 \(\times\) 5x + 4x \(\div\) 2 = 17x.

WORKED_EXAMPLE

> Worked example. Simplify 36y \(\div\) 6 + y + 2 \(\times\) 4y.
>
> No brackets. Division and multiplication come before any addition, so do those first, left to right.
>
> From the left, 36y \(\div\) 6 = 6y. Next, 2 \(\times\) 4y = 8y.
>
> The expression becomes 6y + y + 8y. All are like terms, so add them:
>
> 6y + y + 8y = 15y.
>
> So 36y \(\div\) 6 + y + 2 \(\times\) 4y = 15y.

1. Simplify 4n + 3n \(\times\) 10.
answer: 34n

2. Simplify 32 + 8x \(\div\) 2.
answer: 32 + 4x

3. Simplify 3 \(\times\) 5x + 4x \(\div\) 2.
answer: 17x

4. Simplify 36y \(\div\) 6 + y + 2 \(\times\) 4y.
answer: 15y

Adding and Subtracting Expressions in Brackets

EXPLANATION

When an algebraic expression is written inside brackets, the brackets tell you which terms belong together. To simplify an expression that has brackets, you first remove them — and the key to removing them correctly is reading the sign in front of the bracket.

There are only two rules you need.

Rule 1 — a plus sign in front of the bracket. A `+` before a bracket does not change any of the signs inside. You can simply drop the bracket and write each term as it is.

For example:
$$(3a + 2b) + (5a - b)$$
Remove the brackets, keeping every sign inside exactly as it appears:
$$= 3a + 2b + 5a - b$$
Now collect like terms. The `a`-terms are `3a` and `5a`, and the `b`-terms are `2b` and `-b`:
$$= 8a + b$$

Rule 2 — a minus sign in front of the bracket. A `-` before a bracket changes every sign inside: every `+` becomes `-`, and every `-` becomes `+`. The `-` sign itself is removed along with the bracket.

For example:
$$(9x + 3) - (4x - 2)$$
The first bracket has no sign in front of it (the `9x` is positive), so it stays as it is. The second bracket has `-` in front, so flip every sign inside:
$$= 9x + 3 - 4x + 2$$
Collect the like terms:
$$= 5x + 5$$

Be careful: the `-` in front of the bracket belongs to the whole bracket, not just to the first term inside. Every term inside gets its sign flipped.

WORKED_EXAMPLE

Example 1 — a plus in front of the bracket
Simplify: $(12a - 7) + (4 - 5a)$

Remove both brackets. The first has no sign in front, and the second has a `+` in front, so no sign changes:
$$= 12a - 7 + 4 - 5a$$
Collect the `a`-terms (`12a` and `-5a`) and the number terms (`-7` and `4`):
$$= 7a - 3$$

Example 2 — a minus in front of the bracket
Simplify: $(8x - 2y) - (9x - 4y)$

Remove the first bracket as it is. The second bracket has a `-` in front, so flip every sign inside: `9x` becomes `-9x` and `-4y` becomes `+4y`:
$$= 8x - 2y - 9x + 4y$$
Collect the `x`-terms (`8x` and `-9x`) and the `y`-terms (`-2y` and `4y`):
$$= -x + 2y$$

Example 3 — two minus signs to flip
Simplify: $12 - (7 - 2f) - (3 - f)$

The `12` stands alone and stays as it is. The first bracket has `-` in front, so `7` becomes `-7` and `-2f` becomes `+2f`. The second bracket also has `-` in front, so `3` becomes `-3` and `-f` becomes `+f`:
$$= 12 - 7 + 2f - 3 + f$$
Collect the number terms (`12 - 7 - 3 = 2`) and the `f`-terms (`2f + f = 3f`):
$$= 3f + 2$$

1. Simplify: $(7p + 3) + (5p - 2)$
answer: $12p + 1$

2. Simplify: $(6m - 4) - (2m + 3)$
answer: $4m - 7$

3. Simplify: $5a - (3a - 2b) + (4b - a)$
answer: $a + 6b$

Objective 3

Turning Word Problems into Algebra

EXPLANATION

A word problem gives you information in sentences, and your job is to turn those sentences into an algebraic expression. The key is to look for words that tell you which operation to use.

- "more than" means add (+)
- "less than" means subtract (–)
- "twice" or "2 times" means multiply by 2
- "half of" means divide by 2, or multiply by ½

When an item's cost is compared to another item's cost, start with the item the sentence compares to. For example, "a crayon costs ₦20 less than a pen" means you take the pen's cost and subtract 20. Once you have written the cost of each item as an algebraic term, you can add them together for a total.

WORKED_EXAMPLE

Worked example 1
A pen costs h naira, and a crayon costs ₦20 less than a pen.

(a) Find the cost of a crayon.
"₦20 less than a pen" means we take the pen's cost h and subtract 20.
So the crayon costs h – 20 naira.

(b) Find the cost of a crayon and a pen together.
Add the pen's cost and the crayon's cost:
h + (h – 20)
= h + h – 20
= 2h – 20 naira.

WORKED_EXAMPLE

Worked example 2
A pack of envelopes costs d naira. Four notebooks cost ₦20 more than a pack of envelopes.

Find the total cost of a pack of envelopes and four notebooks.

First, find the cost of the four notebooks. "₦20 more than a pack of envelopes" means we take the pack's cost d and add 20.
So the four notebooks cost d + 20 naira.

Now add the pack of envelopes and the four notebooks:
d + (d + 20)
= d + d + 20
= 2d + 20 naira.

The total cost is 2d + 20 naira.

1. A book costs k naira, and a pen costs ₦25 less than the book. (a) Write an expression for the cost of the pen. (b) Write an expression for the total cost of the pen and the book together, then simplify it.

answer: (a) The pen costs k – 25 naira. (b) Total = k + (k – 25) = 2k – 25 naira.

2. A pencil costs x naira, and an eraser costs ₦15 more than a pencil. (a) Write an expression for the cost of the eraser. (b) Write and simplify an expression for the total cost of two pencils and one eraser.

answer: (a) The eraser costs x + 15 naira. (b) Total = 2x + (x + 15) = 3x + 15 naira.

3. A pen and a pencil together cost x naira. The pen alone costs y naira. (a) Write an expression for the cost of one pencil. (b) Write an expression for the cost of 3 pencils, then simplify it.

answer: (a) One pencil costs x – y naira. (b) 3 pencils cost 3(x – y) = 3x – 3y naira.

C2 `C1 plus the lesson note as source evidence.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Simplification of Algebraic Terms

`subtopic`

Simplification of algebraic expressions: order of operations, brackets and word problems

`term`

Second Term

`week`

9

`lesson_objectives`

• carry out order of operation in algebra
• insert/remove bracket and simplify expression
• solve word problems involving algebraic expression

`entry_behaviour`

• Addition and subtraction of algebraic terms
• Multiplication and division of algebraic terms

`instructional_materials`

• Chart showing algebraic expression.

Objective 1 — carry out order of operation in algebra

objective_text

carry out order of operation in algebra

current_step

1

instructional_events

- provide_learning_guidance
- elicit_performance
- provide_feedback

student_tasks

- 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$

assessment_points

- Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$
- Simplify $7 \times 3a - (3a + 5a) \times 2$

lesson_content


```
[
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  "excerpt": "#### Example 1. Find the value of the following\n\n(a) $18 − 6 \\times 2$\n\n(b) $6
− 18 \\div 3$\n\n(c) $6 \\times 2 − 2 − 40 \\div 4$\n\n**Solution**\n\n(a) $18 − 6 \\times 2$
\n\n$= 18 − 12$\n\n$= 6$\n\n(b) $6 − 18 \\div 3$\n\n$= 6 − 6$\n\n$= 0$\n\n(c) $(6 \\times 2) − 2
− (40 \\div 4)$\n\n$= 12 − 2 − 10$\n\n$= 10 − 10$\n\n$= 0$\n\n#### Example 2. Simplify the
following\n\n(a) $3x \\times 2 + 5x$\n\n(b) $17y − 5 \\times 2y$\n\n**Solution**\n\n(a) $3x \
\times 2 + 5x$\n\n$= 6x + 5x$\n\n$= 11x$\n\n(b) $17y − 5 \\times 2y$\n\n$= 17y − 10y$\n\n$= 7y$
\n\n#### Example 3. Simplify the following\n\n(a) $7x \\div x + 5$\n\n(b) $4x − (6 \\div 3)$
\n\n(c) $(24x \\div 6) + x + (1 \\times 5x)$\n\n**Solution**\n\n(a) $7x \\div x + 5$\n\n$= \
\frac{7x}{x} + 5$\n\n$= 7 + 5$\n\n$= 12$\n\n(b) $4x − (6 \\div 3)$\n\n$= 4x − 2$\n\n(c) $(24x \
\div 6) + x + (1 \\times 5x)$\n\n$= 4x + x + 5x$\n\n$= 5x + 5x$\n\n$= 10x$\n\n#### Activity.
Simplify the following\n\n1. $4n + 3n \\times 10$\n2. $32 + 8x \\div 2$\n3. $3 \\times 5x + 4x \
\div 2$\n4. $36y \\div 6 + y + 2 \\times 4y$"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — insert/remove bracket and simplify expression

objective_text

insert/remove bracket and simplify expression

current_step

2

instructional_events

• provide_learning_guidance
• elicit_performance
• provide_feedback

student_tasks

• (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$

assessment_points

• Simplify $12 - (7 - 2f) - (3 - f)$
• Simplify $(8a - 5) + (14 - 3a)$

lesson_content


```
[
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  "excerpt": "**Hints**\n\n- Always try to simplify the terms inside bracket first.\n- A $+$ sign
before a bracket does not change the signs of the terms inside the bracket, when\n  bracket is
removed.\n- A $-$ sign before a bracket will change all the $+$ sign inside the bracket to $-$
and also\n  change all the $-$ sign inside the bracket to $+$ when the bracket is removed.
\n\n#### Example 1. Simplify\n\n(a) $a + (b + c)$\n\n(b) $a + (b - c)$\n\n(c) $2a - (3b - c)$
\n\n**Solution**\n\n(a) $a + (b + c) = a + b + c$\n\n(b) $a + (b - c) = a + b - c$\n\n(c) $2a -
(3b - c) = 2a - 3b + c$\n\n#### Example 2. Simplify\n\n(a) $a + (a - b)$\n\n(b) $6x + (3 - x)$
\n\n(c) $(3a + 2) + (5a + a)$\n\n**Solution**\n\n(a) $a + (a - b)$\n\n$= a + a - b$\n\n$= 2a - b$
\n\n(b) $6x + (3 - x)$\n\n$= 6x + 3 - x$\n\n$= 6x - x + 3$\n\n$= 5x + 3$\n\n(c) $(3a + 2) + (5a +
a)$\n\n$= 3a + 2 + 5a + a$\n\n$= 3a + 5a + a + 2$\n\n$= 9a + 2$\n\n#### Example 3. Simplify the
following\n\n(a) $5x - (3y + z)$\n\n(b) $10d - (9c - 4d)$\n\n(c) $(6x - y) - (4x - 5y)$
\n\n**Solution**\n\n(a) $5x - (3y + z)$\n\n$= 5x - 3y - z$\n\n(b) $10d - (9c - 4d)$\n\n$= 10d -
9c + 4d$\n\n$= 10d + 4d - 9c$\n\n$= 14d - 9c$\n\n(c) $(6x - y) - (4x - 5y)$\n\n$= 6x - y - 4x +
5y$\n\n$= 6x - 4x + 5y - y$\n\n$= 2x + 4y$\n\n$= 2(x + 2y)$\n\n#### Students activity\n\n(a)
$(12a - 7) + (4 - 5a)$\n\n(b) $(8x - 2y) - (9x - 4y)$"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 3 — solve word problems involving algebraic expression

objective_text

solve word problems involving algebraic expression

current_step

3

instructional_events

• provide_learning_guidance
• elicit_performance
• provide_feedback

student_tasks

• 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.

assessment_points

• A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

lesson_content


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\n\n(a) Find the cost of a pencil\n\n(b) Find the cost of a pen and a
pencil\n\n**Solution**\n\nCost of a pen $= ₦P$\n\n(a) cost of a pencil $= ₦(P − 15)$\n\n(b) So
cost of a pen and a pencil $= ₦P + ₦(P − 15)$\n\n$= ₦P + ₦P − ₦15$\n\n$= ₦2P − ₦15$\n\n$= ₦(2P −
15)$\n\n#### Example 2\n\nA pencil cost $x$ naira and a rubber cost 40 naira more than a pencil.
\n\n(a) find the cost of a rubber\n\n(b) find the cost of a pencil and a
rubber\n\n**Solution**\n\n(a) cost of a pencil $= ₦x$\n\n$\\therefore$ cost of a rubber $= ₦(x +
40)$\n\n(b) cost of a pencil and a rubber:\n\n$= ₦x + ₦(x + 40)$\n\n$= ₦(x + x + 40)$\n\n$= ₦(2x
+ 40)$\n\n#### Example 3\n\nIn a class, there are $x$ boys and $y$ girls last semester. This
semester, 5 boys join and\n3 girls leave the class. How many students are now in this class?
\n\n**Solution**\n\nTotal no $=$ number of boys $+$ number of girls\n\nNumber of boys $= x + 5$
\n\nNumber of girls $= y − 3$\n\nTotal number $= x + 5 + y − 3$\n\n$= x + y + 5 − 3$\n\n$= x + y
+ 2$\n\n#### Student activity\n\n1. A pen cost $h$ naira and a crayon cost ₦20 less than a pen.
\n   (a) Find the cost of a crayon\n   (b) Find the cost of a crayon and a pen\n2. Envelopes cost
$d$ naira per pack, 4 note pad cost 20 naira more than a pack of envelopes.\n   Find the total
cost of a pack of envelope and notepad."
 }
]
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Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

The Order of Operations in Algebra

EXPLANATION

When an algebraic expression contains several different operations, you cannot just work from left to right. There is a fixed order to follow, and the same order works whether the terms carry numbers, letters or both.

Work in this order:

1. **Brackets** — do anything inside brackets first.
2. **Division and Multiplication** — do these next, working from left to right.
3. **Addition and Subtraction** — do these last, again working from left to right.

So in $4n + 3n \times 10$, you multiply before you add: $3n \times 10 = 30n$, then $4n + 30n = 34n$. The n just rides along while you follow the order of the operations.

WORKED_EXAMPLE

Example 1. Simplify $3x \times 2 + 5x$.

Multiplication comes before addition, so handle $3x \times 2$ first:

$3x \times 2 = 6x$

Then add the remaining term:

$6x + 5x = 11x$

So $3x \times 2 + 5x = 11x$.

Example 2. Simplify $17y - 5 \times 2y$.

Multiplication first:

$5 \times 2y = 10y$

Then subtract:

$17y - 10y = 7y$

So $17y - 5 \times 2y = 7y$.

Example 3. Simplify $4x - (6 \div 3)$.

Brackets come first:

$6 \div 3 = 2$

Then subtract:

$4x - 2$

Since $4x$ and 2 are not like terms, the expression is simply left as $4x - 2$.

Example 4. Simplify $(24x \div 6) + x + (1 \times 5x)$.

Work inside each bracket first:

$24x \div 6 = 4x$

$1 \times 5x = 5x$

Now add the like terms:

$4x + x + 5x = 10x$

So $(24x \div 6) + x + (1 \times 5x) = 10x$.

EXPLANATION

Division works the same way as multiplication in the order. In $7x \div x + 5$, you divide before you add:

$7x \div x = \frac{7x}{x} = 7$

Then $7 + 5 = 12$.

Watch out: the division applies only to the term it touches. In $32 + 8x \div 2$, only the $8x$ is divided by 2, so it becomes $4x$, and the expression is $32 + 4x$.

1. Simplify $4n + 3n \times 10$.
answer: $34n$

2. Simplify $32 + 8x \div 2$.
answer: $32 + 4x$

3. Simplify $3 \times 5x + 4x \div 2$.
answer: $17x$

4. Simplify $36y \div 6 + y + 2 \times 4y$.
answer: $15y$

Inserting and Removing Brackets

EXPLANATION
When an algebraic expression contains brackets, you often need to remove them before you can collect like terms and simplify. Here are the rules to follow.

First, always try to simplify the terms **inside** the bracket before you remove it.

Second, a **plus** sign (+) in front of a bracket does not change the signs of the terms inside the bracket when you remove it.

Third, a **minus** sign (–) in front of a bracket changes every + inside the bracket to –, and every – inside the bracket to +, when you remove it.

So removing a bracket is really a small sign-change job: keep the signs when a + sits in front, and flip every sign when a – sits in front.

WORKED_EXAMPLE
Example 1. Simplify
(a) $a + (b + c)$ (b) $a + (b - c)$ (c) $2a - (3b - c)$

Solution
(a) The sign before the bracket is +, so the signs inside stay as they are:
$$a + (b + c) = a + b + c$$

(b) Again the + before the bracket means nothing changes:
$$a + (b - c) = a + b - c$$

(c) Here a – stands before the bracket, so every sign inside flips: the + before $3b$ becomes –, and the – before c becomes +:
$$2a - (3b - c) = 2a - 3b + c$$

WORKED_EXAMPLE
Example 2. Simplify
(a) $a + (a - b)$ (b) $6x + (3 - x)$ (c) $(3a + 2) + (5a + a)$

Solution
(a) The + before the bracket keeps the signs inside, then collect like terms:
$$a + (a - b) = a + a - b = 2a - b$$

(b) Remove the bracket (the + before it changes nothing), then rearrange so like terms sit together:
$$6x + (3 - x) = 6x + 3 - x = 6x - x + 3 = 5x + 3$$

(c) Here we have two brackets joined by +. First simplify inside each bracket: $(3a + 2)$ stays as it is, and $(5a + a) = 6a$. So
$$(3a + 2) + (5a + a) = 3a + 2 + 5a + a = 3a + 5a + a + 2 = 9a + 2$$

WORKED_EXAMPLE
Example 3. Simplify
(a) $5x - (3y + z)$ (b) $10d - (9c - 4d)$ (c) $(6x - y) - (4x - 5y)$

Solution
(a) A – before the bracket flips every sign inside:
$$5x - (3y + z) = 5x - 3y - z$$

(b) Flip every sign inside the bracket, then collect like terms:
$$10d - (9c - 4d) = 10d - 9c + 4d = 10d + 4d - 9c = 14d - 9c$$

(c) The – before the second bracket flips both of its signs, and the first bracket keeps its signs (nothing in front of it). Then collect like terms:
$$(6x - y) - (4x - 5y) = 6x - y - 4x + 5y = 6x - 4x + 5y - y = 2x + 4y = 2(x + 2y)$$

1. Simplify $(12a - 7) + (4 - 5a)$.
answer: $7a - 3$

2. Simplify $(8x - 2y) - (9x - 4y)$.
answer: $-x + 2y$

3. Simplify $12 - (7 - 2f) - (3 - f)$.
answer: $2 + 3f$

4. Simplify $(8a - 5) + (14 - 3a)$.
answer: $5a + 9$

Objective 3

Solving Word Problems with Algebraic Expressions

EXPLANATION

A word problem hides an algebraic expression inside ordinary language. To solve it, read the problem, decide what letter stands for the unknown amount, and then translate each sentence into algebra. Words like less than tell you to subtract, more than tells you to add, and altogether or total tells you to add the parts together. Once you have written the expression, simplify it by collecting like terms as you have already learnt.

WORKED_EXAMPLE

Example 1
A pen costs ₦P and a pencil costs ₦15 less than a pen.
(a) Find the cost of a pencil.
(b) Find the cost of a pen and a pencil.

Solution
Cost of a pen = ₦P
(a) The pencil costs ₦15 less than the pen, so:
Cost of a pencil = ₦(P – 15)
(b) Cost of a pen and a pencil = ₦P + ₦(P – 15)
= ₦P + ₦P – ₦15
= ₦2P – ₦15
= ₦(2P – 15)

WORKED_EXAMPLE

Example 2
A pencil costs ₦x and a rubber costs ₦40 more than a pencil.
(a) Find the cost of a rubber.
(b) Find the cost of a pencil and a rubber.

Solution
Cost of a pencil = ₦x
(a) The rubber costs ₦40 more, so:
Cost of a rubber = ₦(x + 40)
(b) Cost of a pencil and a rubber = ₦x + ₦(x + 40)
= ₦(x + x + 40)
= ₦(2x + 40)

WORKED_EXAMPLE

Example 3
In a class there are x boys and y girls. This semester, 5 boys join and 3 girls leave the class. How many students are now in the class?

Solution
Total number = number of boys + number of girls
Number of boys = x + 5
Number of girls = y – 3
Total number = x + 5 + y – 3
= x + y + 5 – 3
= x + y + 2
So there are now (x + y + 2) students in the class.

1. A pen costs ₦h and a crayon costs ₦20 less than a pen.
(a) Find the cost of a crayon.
(b) Find the cost of a crayon and a pen.

answer: (a) ₦(h – 20)
(b) ₦h + ₦(h – 20) = ₦(2h – 20)

2. A pack of envelopes costs ₦d, and a notepad costs ₦20 more than a pack of envelopes. Find the total cost of a pack of envelopes and a notepad.

answer: ₦(2d + 20)

3. A pen and a pencil altogether cost ₦x. If a pen costs ₦y, find the cost of 3 pencils.

answer: ₦(3x – 3y) or ₦3(x – y)

C3 `C2 plus a validation and repair pass over the draft.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Simplification of Algebraic Terms

`subtopic`

Simplification of algebraic expressions: order of operations, brackets and word problems

`term`

Second Term

`week`

9

`lesson_objectives`

• carry out order of operation in algebra
• insert/remove bracket and simplify expression
• solve word problems involving algebraic expression

`entry_behaviour`

• Addition and subtraction of algebraic terms
• Multiplication and division of algebraic terms

`instructional_materials`

• Chart showing algebraic expression.

Objective 1 — carry out order of operation in algebra

objective_text

carry out order of operation in algebra

current_step

1

instructional_events

- provide_learning_guidance
- elicit_performance
- provide_feedback

student_tasks

- 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$

assessment_points

- Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$
- Simplify $7 \times 3a - (3a + 5a) \times 2$

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Example 1. Find the value of the following\n\n(a) $18 − 6 \\times 2$\n\n(b) $6
− 18 \\div 3$\n\n(c) $6 \\times 2 − 2 − 40 \\div 4$\n\n**Solution**\n\n(a) $18 − 6 \\times 2$
\n\n$= 18 − 12$\n\n$= 6$\n\n(b) $6 − 18 \\div 3$\n\n$= 6 − 6$\n\n$= 0$\n\n(c) $(6 \\times 2) − 2
− (40 \\div 4)$\n\n$= 12 − 2 − 10$\n\n$= 10 − 10$\n\n$= 0$\n\n#### Example 2. Simplify the
following\n\n(a) $3x \\times 2 + 5x$\n\n(b) $17y − 5 \\times 2y$\n\n**Solution**\n\n(a) $3x \
\times 2 + 5x$\n\n$= 6x + 5x$\n\n$= 11x$\n\n(b) $17y − 5 \\times 2y$\n\n$= 17y − 10y$\n\n$= 7y$
\n\n#### Example 3. Simplify the following\n\n(a) $7x \\div x + 5$\n\n(b) $4x − (6 \\div 3)$
\n\n(c) $(24x \\div 6) + x + (1 \\times 5x)$\n\n**Solution**\n\n(a) $7x \\div x + 5$\n\n$= \
\frac{7x}{x} + 5$\n\n$= 7 + 5$\n\n$= 12$\n\n(b) $4x − (6 \\div 3)$\n\n$= 4x − 2$\n\n(c) $(24x \
\div 6) + x + (1 \\times 5x)$\n\n$= 4x + x + 5x$\n\n$= 5x + 5x$\n\n$= 10x$\n\n#### Activity.
Simplify the following\n\n1. $4n + 3n \\times 10$\n2. $32 + 8x \\div 2$\n3. $3 \\times 5x + 4x \
\div 2$\n4. $36y \\div 6 + y + 2 \\times 4y$"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — insert/remove bracket and simplify expression

objective_text

insert/remove bracket and simplify expression

current_step

2

instructional_events

• provide_learning_guidance
• elicit_performance
• provide_feedback

student_tasks

• (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$

assessment_points

• Simplify $12 - (7 - 2f) - (3 - f)$
• Simplify $(8a - 5) + (14 - 3a)$

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "**Hints**\n\n- Always try to simplify the terms inside bracket first.\n- A $+$ sign
before a bracket does not change the signs of the terms inside the bracket, when\n  bracket is
removed.\n- A $-$ sign before a bracket will change all the $+$ sign inside the bracket to $-$
and also\n  change all the $-$ sign inside the bracket to $+$ when the bracket is removed.
\n\n#### Example 1. Simplify\n\n(a) $a + (b + c)$\n\n(b) $a + (b - c)$\n\n(c) $2a - (3b - c)$
\n\n**Solution**\n\n(a) $a + (b + c) = a + b + c$\n\n(b) $a + (b - c) = a + b - c$\n\n(c) $2a -
(3b - c) = 2a - 3b + c$\n\n#### Example 2. Simplify\n\n(a) $a + (a - b)$\n\n(b) $6x + (3 - x)$
\n\n(c) $(3a + 2) + (5a + a)$\n\n**Solution**\n\n(a) $a + (a - b)$\n\n$= a + a - b$\n\n$= 2a - b$
\n\n(b) $6x + (3 - x)$\n\n$= 6x + 3 - x$\n\n$= 6x - x + 3$\n\n$= 5x + 3$\n\n(c) $(3a + 2) + (5a +
a)$\n\n$= 3a + 2 + 5a + a$\n\n$= 3a + 5a + a + 2$\n\n$= 9a + 2$\n\n#### Example 3. Simplify the
following\n\n(a) $5x - (3y + z)$\n\n(b) $10d - (9c - 4d)$\n\n(c) $(6x - y) - (4x - 5y)$
\n\n**Solution**\n\n(a) $5x - (3y + z)$\n\n$= 5x - 3y - z$\n\n(b) $10d - (9c - 4d)$\n\n$= 10d -
9c + 4d$\n\n$= 10d + 4d - 9c$\n\n$= 14d - 9c$\n\n(c) $(6x - y) - (4x - 5y)$\n\n$= 6x - y - 4x +
5y$\n\n$= 6x - 4x + 5y - y$\n\n$= 2x + 4y$\n\n$= 2(x + 2y)$\n\n#### Students activity\n\n(a)
$(12a - 7) + (4 - 5a)$\n\n(b) $(8x - 2y) - (9x - 4y)$"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 3 — solve word problems involving algebraic expression

objective_text

solve word problems involving algebraic expression

current_step

3

instructional_events

• provide_learning_guidance
• elicit_performance
• provide_feedback

student_tasks

• 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.

assessment_points

• A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Example 1\n\nA pen cost $P$ naira and a pencil cost 15 naira less than a pen.
\n\n(a) Find the cost of a pencil\n\n(b) Find the cost of a pen and a
pencil\n\n**Solution**\n\nCost of a pen $= ₦P$\n\n(a) cost of a pencil $= ₦(P − 15)$\n\n(b) So
cost of a pen and a pencil $= ₦P + ₦(P − 15)$\n\n$= ₦P + ₦P − ₦15$\n\n$= ₦2P − ₦15$\n\n$= ₦(2P −
15)$\n\n#### Example 2\n\nA pencil cost $x$ naira and a rubber cost 40 naira more than a pencil.
\n\n(a) find the cost of a rubber\n\n(b) find the cost of a pencil and a
rubber\n\n**Solution**\n\n(a) cost of a pencil $= ₦x$\n\n$\\therefore$ cost of a rubber $= ₦(x +
40)$\n\n(b) cost of a pencil and a rubber:\n\n$= ₦x + ₦(x + 40)$\n\n$= ₦(x + x + 40)$\n\n$= ₦(2x
+ 40)$\n\n#### Example 3\n\nIn a class, there are $x$ boys and $y$ girls last semester. This
semester, 5 boys join and\n3 girls leave the class. How many students are now in this class?
\n\n**Solution**\n\nTotal no $=$ number of boys $+$ number of girls\n\nNumber of boys $= x + 5$
\n\nNumber of girls $= y − 3$\n\nTotal number $= x + 5 + y − 3$\n\n$= x + y + 5 − 3$\n\n$= x + y
+ 2$\n\n#### Student activity\n\n1. A pen cost $h$ naira and a crayon cost ₦20 less than a pen.
\n   (a) Find the cost of a crayon\n   (b) Find the cost of a crayon and a pen\n2. Envelopes cost
$d$ naira per pack, 4 note pad cost 20 naira more than a pack of envelopes.\n   Find the total
cost of a pack of envelope and notepad."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Carrying Out the Order of Operations in Algebra

EXPLANATION

When an algebraic expression contains more than one operation, you cannot work from left to right. Some operations must be done before others. The rule is:

Multiply and divide first, then add and subtract.

This is called the order of operations. So in any expression, do every multiplication and every division before you tackle any addition or subtraction. Once all the multiplying and dividing is done, you add and subtract the terms that remain, using what you already know about adding and subtracting like terms.

WORKED_EXAMPLE

Example 1. Find the value of each of the following.

(a) $18 - 6 \times 2$

The multiplication must be done before the subtraction.

$18 - 6 \times 2 = 18 - 12 = 6$

(b) $6 - 18 \div 3$

The division must be done before the subtraction.

$6 - 18 \div 3 = 6 - 6 = 0$

(c) $6 \times 2 - 2 - 40 \div 4$

Do the multiplication and the division first.

$6 \times 2 - 2 - 40 \div 4 = 12 - 2 - 10$

Now subtract from left to right.

$12 - 2 - 10 = 10 - 10 = 0$

WORKED_EXAMPLE

Example 2. Simplify each of the following.

(a) $3x \times 2 + 5x$

Do the multiplication first.

$3x \times 2 + 5x = 6x + 5x = 11x$

(b) $17y - 5 \times 2y$

Do the multiplication first.

$17y - 5 \times 2y = 17y - 10y = 7y$

WORKED_EXAMPLE

Example 3. Simplify each of the following.

(a) $7x \div x + 5$

Do the division first.

$7x \div x + 5 = \frac{7x}{x} + 5 = 7 + 5 = 12$

(b) $4x - (6 \div 3)$

The brackets tell you what to do first. Work out the inside of the brackets.

$4x - (6 \div 3) = 4x - 2$

(c) $(24x \div 6) + x + (1 \times 5x)$

Work inside each pair of brackets first.

$(24x \div 6) + x + (1 \times 5x) = 4x + x + 5x$

Now add the like terms.

$4x + x + 5x = 5x + 5x = 10x$

1. Simplify $4n + 3n \times 10$.
answer: $34n$

2. Simplify $32 + 8x \div 2$.
answer: $32 + 4x$

3. Simplify $3 \times 5x + 4x \div 2$.
answer: $17x$

4. Simplify $36y \div 6 + y + 2 \times 4y$.
answer: $15y$

Removing Brackets and Simplifying

EXPLANATION

When an algebraic expression has brackets in it, you often want to remove the brackets so you can gather the like terms together. Here are the two rules for removing a bracket.

- If there is a **+** sign in front of the bracket, removing the bracket **does not change** any sign of the terms inside. They stay exactly as they are.
- If there is a **–** sign in front of the bracket, removing the bracket **changes every sign** inside: every + becomes – and every – becomes +.

You can see this with small letters first.

Take $a + (b + c)$. The + in front of the bracket means the terms inside keep their signs, so:
$$a + (b + c) = a + b + c$$

Now $a + (b - c)$. Still a + in front, so the – stays a –:
$$a + (b - c) = a + b - c$$

But $2a - (3b - c)$ has a **–** in front of the bracket. That – changes both signs inside: $3b$ (a +) becomes $-3b$, and $-c$ becomes $+c$:
$$2a - (3b - c) = 2a - 3b + c$$

Once the brackets are gone, finish by collecting the like terms and adding or subtracting them.

WORKED_EXAMPLE

Let's work through a few examples using these rules.

Example 1. Simplify $a + (a - b)$.

Remove the bracket. There is a + in front, so the terms inside keep their signs:
$$a + a - b$$
Now collect the like terms:
$$= 2a - b$$

Example 2. Simplify $6x + (3 - x)$.

Remove the bracket (the + keeps the signs):
$$= 6x + 3 - x$$
Gather the x terms together:
$$= 6x - x + 3$$
$$= 5x + 3$$

Example 3. Simplify $5x - (3y + z)$.

There is a – in front, so every sign inside changes: $3y$ becomes $-3y$ and z becomes $-z$:
$$= 5x - 3y - z$$
There are no like terms to collect, so this is the answer.

Example 4. Simplify $(6x - y) - (4x - 5y)$.

The first bracket has no sign in front, so it comes down unchanged:
$$6x - y$$
The second bracket has a – in front, so both signs inside change: $4x$ becomes $-4x$ and $-5y$ becomes $+5y$:
$$= 6x - y - 4x + 5y$$
Now collect like terms:
$$= 6x - 4x + 5y - y$$
$$= 2x + 4y$$

> Notice how the minus in front of a bracket is what forces you to change the signs — a plus sign leaves them alone.

1. Simplify $(12a - 7) + (4 - 5a)$.
answer: $7a - 3$

2. Simplify $(8x - 2y) - (9x - 4y)$.
answer: $-x + 2y$

3. Simplify $12 - (7 - 2f) - (3 - f)$.
answer: $2 + 3f$

4. Simplify $(8a - 5) + (14 - 3a)$.
answer: $5a + 9$

Turning Words into Algebra: Word Problems

EXPLANATION

A word problem describes a situation in words, and your job is to turn those words into an algebraic expression. The words tell you which operation to use.

The key phrases to watch for:

- "less than" means subtract. If a pencil costs 15 naira less than a pen, take 15 away from the pen's cost.
- "more than" means add. If a rubber costs 40 naira more than a pencil, add 40 to the pencil's cost.
- "altogether", "total", or "and" (when joining costs) means add the two amounts.

Start by writing down what each letter stands for. Then translate each sentence into algebra, one step at a time, and finally simplify your expression by collecting like terms.

Let's see this in action.

WORKED_EXAMPLE

Example 1: "less than"

A pen costs ₦P and a pencil costs 15 naira less than a pen.

(a) Find the cost of a pencil.

"Less than a pen" means we take 15 away from the cost of a pen. Since a pen costs ₦P, the pencil costs:

₦(P – 15)

(b) Find the cost of a pen and a pencil.

The total is the cost of the pen plus the cost of the pencil:

₦P + ₦(P – 15)

Now simplify. First remove the bracket:

₦P + ₦P – ₦15 = ₦2P – ₦15

So the pen and pencil together cost ₦(2P – 15).

WORKED_EXAMPLE

Example 2: "more than"

A pencil costs ₦x and a rubber costs 40 naira more than a pencil.

(a) Find the cost of a rubber.

"More than a pencil" means we add 40 to the cost of the pencil. Since the pencil costs ₦x, the rubber costs:

₦(x + 40)

(b) Find the cost of a pencil and a rubber.

Add the two costs together:

₦x + ₦(x + 40)

Remove the bracket:

= ₦(x + x + 40) = ₦(2x + 40)

So the pencil and rubber together cost ₦(2x + 40).

WORKED_EXAMPLE
Example 3: changing numbers of people

In a class there are x boys and y girls last semester. This semester, 5 boys join and 3 girls leave the class. How many students are now in the class?

Think about each group separately.

Number of boys now = x + 5 (5 boys join, so add 5)
Number of girls now = y – 3 (3 girls leave, so subtract 3)

The total number of students is the boys plus the girls:

Total = x + 5 + y – 3

Collect the numbers together:

= x + y + 5 – 3 = x + y + 2

So the class now has x + y + 2 students.

1. 1. A pen costs ₦h and a crayon costs ₦20 less than a pen. (a) Find the cost of a crayon. (b) Find the cost of a crayon and a pen.
answer: (a) ₦(h – 20) (b) ₦(2h – 20)

2. 2. Envelopes cost ₦d per pack, and a notepad costs 20 naira more than a pack of envelopes. Find the total cost of a pack of envelopes and a notepad.
answer: ₦(2d + 20)

3. 3. A pen and a pencil altogether cost ₦x. If the pen costs ₦y, find the cost of 3 pencils.
answer: ₦(3x – 3y)

Result — per rubric dimension

Scores are 1–5. Every figure is the mean of 3 draws rated by 2 judge pass(es) — 6 judgments per cell.

Rubric	C0	C1	C2	C3	C1 – C0	C2 – C1	C3 – C2
Objective and curriculum alignment	4.00	5.00	5.00	5.00	**+1.00**	+0.00	+0.00
Taught-lesson alignment	3.00	4.83	5.00	5.00	**+1.83**	**+0.17**	+0.00
Scope and conceptual correctness	3.67	4.83	4.33	4.00	**+1.17**	**-0.50**	**-0.33**
Student-facing clarity and usefulness	4.50	5.00	5.00	5.00	**+0.50**	+0.00	+0.00

Why the judge scored it that way

Every rationale the judge wrote for this lesson, deduplicated across draws and passes. These are its words, not a summary.

Objective and curriculum alignment — C0 4.00, C1 5.00, C2 5.00, C3 5.00

- `C0` All three lesson objectives are addressed in dedicated sections, and the curriculum target for JSS1 Second Term algebra is well served. One minor overstep (solving an equation rather than just writing an expression) slightly exceeds the stated objective of writing/simplifying algebraic expressions, preventing a perfect 5.
- `C0` All three lesson objectives are addressed clearly and in reasonable depth. The pack loses one point because the order-of-operations section does not include examples with mixed multiplication and division of algebraic terms (e.g., the type in the lesson's set questions like 4n + 3n × 10 or 36y ÷ 6 + y + 2 × 4y), staying closer to purely numerical BODMAS or simple expansions rather than the algebraic BODMAS the lesson specifically targets.
- `C0` All three lesson objectives are addressed with worked examples and practice questions in each section. The inclusion of factorisation is a minor addition beyond the stated objectives but does not detract from the core coverage.
- `C0` The pack covers all three stated objectives in an organised way. It loses one mark because the word-problem section goes slightly beyond the lesson's scope by introducing solving equations (finding a numerical answer), whereas the lesson objectives focused on forming and simplifying expressions.
- `C0` All three stated objectives are explicitly addressed with explanations, worked examples, and practice questions aligned to JSS1 second-term algebra. The pack loses one point because it does not treat the order-of-operations section as deeply as the lesson's own class exercises (e.g., multi-step expressions like 36y ÷ 6 + y + 2 × 4y are absent from the revision practice).
- `C0` The pack covers all three stated objectives clearly and in a way consistent with JSS1 second-term algebra. The minor drift toward solving equations (rather than purely writing and simplifying expressions) slightly exceeds the stated objectives but does not undermine overall alignment.
- `C1` The revision pack covers every stated objective in the correct curriculum sequence for JSS1 Second Term Week 9, with no objectives omitted or added from outside the lesson scope.
- `C1` The revision pack maps one-to-one onto all three stated lesson objectives and uses the same curriculum language (BODMAS) expected at JSS1 Second Term. Nothing from the objectives is omitted.
- `C1` The pack addresses every stated lesson objective in the correct JSS1 Second Term context, with no objective left out and no extraneous curriculum content introduced.
- `C1` The revision pack covers every stated objective fully and in the correct JSS1 Second Term sequence. There are no gaps and no objectives addressed that fall outside the lesson scope.
- `C1` The pack covers every stated objective in the lesson plan with matching content and appropriate JSS1 level. There is complete alignment with the curriculum target for JSS1 Term 2 Week 9 algebraic simplification.
- `C1` The pack covers every stated objective in the correct JSS1 scope — order of operations, bracket manipulation, and algebraic word problems — with no gaps and no extraneous content. It maps directly onto what the lesson plan specifies students should learn.
- `C2` The revision pack addresses every stated objective in the correct sequence, and includes both the class practice questions and the teacher's evaluation questions as practice items, demonstrating complete curriculum alignment for this JSS1 second-term topic.

- `C2` The pack covers every stated objective in the lesson plan with appropriate examples and practice questions, and also incorporates the teacher's own evaluation questions, demonstrating full alignment with both the lesson objectives and JSS1 second-term curriculum expectations.
- `C2` The pack covers all three stated objectives in the correct JSS1 Second Term Week 9 curriculum context, with practice questions drawn directly from both the lesson set and the teacher's evaluation. Alignment is thorough and complete.
- `C2` The pack covers every stated objective in the correct JSS1 Second Term scope, with examples and practice tasks that map directly onto each curriculum target. Nothing is omitted and nothing is added beyond what the lesson intended.
- `C2` The pack addresses every stated objective in the lesson plan and replicates the teacher's own set questions and evaluation tasks, demonstrating full alignment with both the JSS1 curriculum target and the lesson's intended learning outcomes.
- `C2` The pack covers every stated objective in the same order they appear in the lesson plan and incorporates both the in-class practice questions and the teacher's evaluation questions, achieving full alignment with the curriculum targets for JSS1 second-term week 9.
- `C3` The pack addresses every stated objective with both explanation and practice, and the questions mirror exactly what the teacher set for both classwork and evaluation. Curriculum alignment is complete.
- `C3` The revision pack comprehensively covers all three stated objectives in the correct JSS1 Second Term scope, leaving no objective unaddressed. Curriculum alignment is complete.
- `C3` Every stated objective of the lesson is addressed in the revision pack, with appropriate examples and practice aligned to the JSS1 second-term curriculum target on simplification of algebraic expressions.
- `C3` The pack addresses every stated objective for the JSS1 second-term week-9 lesson with appropriate depth. There are no gaps and no off-topic material.
- `C3` The revision pack covers every stated lesson objective in the correct JSS1 Second Term scope, with worked examples and practice questions that map directly onto the teacher's own set questions and evaluation tasks.
- `C3` The revision pack covers every stated objective fully and in the correct JSS1 second-term scope. No objective is omitted or distorted, and the curriculum target is cleanly met.

Taught-lesson alignment — C0 3.00, C1 4.83, C2 5.00, C3 5.00

- `C0` The structural sequence is respected and the three topics are covered, but the pack replaces the teacher's specific classroom examples and practice items with new ones, and introduces equation-solving in the word-problem section that was not part of the taught lesson, reducing fidelity to what was actually taught.
- `C0` The pack covers the same broad topics as the lesson but shifts the emphasis and examples in ways that don't fully reflect what was taught: the algebraic order-of-operations examples are weaker than those set in class, and the word problems go beyond the lesson's own scenarios and depth. The sequence is broadly correct but the specific activities and examples diverge noticeably.
- `C0` While the pack follows the same three-topic sequence as the lesson, it diverges in emphasis and depth — the order-of-operations practice is simpler than the teacher's examples, the lesson's multiplication/division evaluation questions are absent, and the word problems shift toward equation-solving rather than expression-building as the teacher intended.
- `C0` The pack follows the same three-topic sequence as the lesson and reflects its general approach, but the depth and type of exercises diverge noticeably from what the teacher actually set — simpler order-of-operations items, more complex bracket work, and equation-solving in word problems that the lesson did not teach.
- `C0` The pack follows the same three-part sequence as the lesson, but its scope in both directions (simpler BODMAS, more advanced bracket expansion, new problem types) means it does not closely mirror what students actually did in class, reducing confidence that a student working alone is reviewing exactly what was taught.
- `C0` While the pack's three sections map loosely onto the lesson's three objectives, the specific examples, problem types, and emphasis differ noticeably from what the teacher actually set and taught; the inclusion of factorisation and equation-solving goes beyond the lesson's scope and sequence.
- `C1` The pack faithfully reflects the lesson's sequence, method, and depth, including the teacher's own evaluation questions as practice; the slight extension on bracket insertion is a minor addition that does not distort the lesson emphasis.
- `C1` The pack faithfully reproduces the lesson's sequence, emphasis, and depth. Every activity type from the lesson — worked examples, independent practice — is represented, and the teacher's own questions are integrated throughout.
- `C1` The revision pack mirrors the lesson's sequence, uses the same examples the teacher demonstrated, and replicates the independent-practice questions, making it a faithful reflection of what was actually taught.
- `C1` The pack faithfully reproduces the lesson's sequence, examples, and emphasis, giving students a direct bridge from what was taught in class to home revision. The depth matches JSS1 expectations and the teacher's intended level.

- `C1` The revision pack faithfully reproduces the lesson's sequence, emphasis, and depth, and its examples closely mirror what the teacher set during the lesson and evaluation. A student working through this pack would be revising exactly what was taught in class.
- `C1` The revision pack faithfully reproduces the lesson's sequence, worked examples, and independent-practice questions, reflecting the teacher's intended method and depth throughout all three sections.
- `C2` The pack mirrors the lesson's structure, method, and specific activities almost exactly — from the initial rule explanation through board-style worked examples to independent practice — reflecting both the sequence and depth the teacher employed.
- `C2` The pack tracks the taught lesson closely in sequence, emphasis, and method, and the practice sets are identical to the teacher's in-class questions, giving a clear reflection of what students actually did during the lesson.
- `C2` The revision pack faithfully reproduces the lesson's sequence, method, and emphasis, and integrates both the in-class practice sets and the teacher's evaluation questions, making it a strong take-home mirror of what was taught.
- `C2` The revision pack is structured to walk the student through the lesson in the order it was taught, using the same questions and contexts, which gives strong continuity between what was taught in class and what is reviewed at home.
- `C2` The pack's structure, sequencing, and choice of tasks map precisely onto the lesson as planned, including the same examples and practice items the teacher intended to use in class, giving strong alignment to how the lesson was taught.
- `C2` The revision pack faithfully reflects the sequence, method, and depth of the taught lesson, including the same exemplar style (step-by-step board work) and the same practice questions students would have seen in class.
- `C3` The revision pack mirrors the lesson's sequence, method, and depth so closely that a student could use it to reconstruct what happened in class; every activity type and example type from the lesson appears in the pack.
- `C3` The pack reflects the lesson's sequence, emphasis, and depth almost perfectly, including the teacher's own practice sets and evaluation questions as revision exercises with full worked solutions.
- `C3` The revision pack faithfully mirrors the lesson's sequence, emphasis, and expected depth, making it a direct continuation of what students experienced in class.
- `C3` The revision pack mirrors the lesson's sequence, emphasis, and the specific questions the teacher used in class, making it a faithful take-home extension of what was taught.
- `C3` The pack faithfully reproduces the lesson's sequence, depth, and student activities; it reads as a direct take-home extension of what the teacher planned to do in class.
- `C3` The pack mirrors the lesson's sequence, method, activities, and depth with high fidelity, including re-use of the teacher's own exercise questions. It reads as a faithful home-study version of what was taught in class.

Scope and conceptual correctness — C0 3.67, C1 4.83, C2 4.33, C3 4.00

- `C0` The mathematics presented is correct throughout, with no sign errors, arithmetic mistakes, or misconceptions introduced. The only concern is a slight scope creep into equation-solving, which is not yet the lesson's focus but does not introduce any errors.
- `C0` The mathematics throughout the pack is accurate and the conceptual explanations (BODMAS ladder, sign-flip rule, collecting like terms) are correct. The pack introduces two-bracket expansion with coefficients (e.g., 2(3x+5)+4(x–1)) which slightly exceeds the depth of the lesson's own bracket examples, but this is a minor scope extension rather than an error, hence a 4 rather than 5.
- `C0` The mathematical content is accurate with no conceptual errors, but the pack exceeds the lesson's scope by introducing factorisation and equation-solving, which are distinct skills beyond simplification of algebraic expressions as planned for this lesson.
- `C0` The mathematics is conceptually sound and contains no errors, but the pack strays outside the lesson scope by including linear equation solving, which is beyond JSS1 week 9 simplification objectives.
- `C0` The mathematics throughout is accurate with no errors in the worked examples or answer key. The only concern is scope creep — expanding brackets with a coefficient belongs to a slightly later stage and was not part of the taught lesson — but it is not incorrect, so the score stays at 4.
- `C0` The mathematics presented is correct throughout, but the pack strays outside the lesson boundary by introducing factorisation and linear equation solving, both of which are beyond what was taught in this lesson.
- `C1` No conceptual or arithmetic errors were found across the entire pack; the content stays squarely within JSS1 algebraic simplification scope and treats every concept accurately.
- `C1` There are no mathematical errors or conceptual misconceptions in the pack. The content stays firmly within the JSS1 scope — no over-extension into equations or factorisation — and every answer in the 'Check your work' section is accurate.

- `C1` Every mathematical statement, worked example, and answer key entry in the pack is correct, and the content stays entirely within the scope of the planned lesson without straying into out-of-scope topics.
- `C1` The mathematics is sound throughout with no conceptual errors. The small extension into bracket-insertion practice is educationally appropriate and not misleading, but it does nudge slightly past the lesson boundary, preventing a perfect 5.
- `C1` The pack stays strictly within the scope of the lesson — no topics beyond the lesson boundary are introduced — and every mathematical statement, rule, and worked solution is conceptually and computationally correct.
- `C1` Every calculation and conceptual statement in the pack is correct, and the material stays firmly within the JSS1 second-term scope without introducing anything beyond what the lesson intended.
- `C2` The pack is mathematically accurate throughout with no conceptual errors; the only mild concern is one example (7x ÷ x) that edges slightly beyond the lesson's apparent depth, but this does not represent a significant scope violation.
- `C2` The conceptual treatment of BODMAS, bracket rules, and word-problem translation is sound and age-appropriate, and all final answers are correct; one minor intermediate-step error in a worked example slightly reduces the score but does not mislead students about the answer.
- `C2` There are no conceptual or computational errors anywhere in the pack, and the content remains appropriately bounded to the lesson's expected depth without straying into higher-level algebra.
- `C2` The mathematics is accurate throughout with no conceptual errors. One example (7x ÷ x) nudges very slightly outside the lesson's stated scope, but it does not introduce an incorrect idea and does not affect the practice questions students are asked to do.
- `C2` Concepts and answers are overwhelmingly correct and appropriately scoped for JSS1; the only issue is a minor notational ambiguity in one worked example that could momentarily confuse a student but does not constitute a mathematical error.
- `C2` Every concept, rule, and worked answer in the pack is mathematically correct, and the content stays within the expected lesson boundary for JSS1 simplification of algebraic expressions without straying into higher-level topics.
- `C3` The mathematics is correct throughout and the pack stays well within JSS1 simplification territory; the single instance of optional factorisation is a very minor over-reach that does not introduce any error.
- `C3` The mathematics throughout is correct and the scope is tightly controlled; the brief introduction of factorisation in one example is a minor overreach but does not introduce any error.
- `C3` The pack is mathematically accurate throughout and stays well within the lesson's scope; the minor extension of variable cancellation in one example is handled correctly and does not introduce any error or confusion.
- `C3` The mathematics throughout the pack is accurate and the content stays largely within lesson boundaries; the single minor scope excursion (factorising) is unlikely to confuse learners and does not constitute a conceptual error.
- `C3` The pack is mathematically accurate throughout with correct answers in the answer key; the single instance of factoring goes marginally beyond the lesson scope but is handled correctly and does not mislead students.
- `C3` The pack stays firmly within the lesson boundary and is mathematically correct throughout. One minor notational imprecision in an intermediate step slightly reduces clarity but does not introduce a conceptual error.

Student-facing clarity and usefulness — C0 4.50, C1 5.00, C2 5.00, C3 5.00

- `C0` The pack is clearly written, logically structured, and practically usable as independent take-home material for JSS1 students; the self-check answers and step-by-step models are strong features, with only the proximity of answers to questions being a minor pedagogical concern.
- `C0` The pack is exceptionally well structured for independent home use: language is accessible to JSS1 level, steps are explicit, and the answer key supports self-assessment. It is one of the clearest take-home revision packs a student could receive for this topic.
- `C0` The pack is generally well-structured and student-friendly, with clear steps, worked examples, and full answer keys that make it suitable for independent home revision. The raw HTML tags are a presentation flaw that slightly reduces usability in a printed format.
- `C0` The pack is exceptionally well structured for independent home use: clear language, generous worked examples, and a detailed answer key make it highly useful for a JSS1 learner revising without teacher support.
- `C0` The pack is exceptionally well-structured for independent home use: language is clear, steps are explicit, examples precede practice, and full solutions are provided. It is one of the most student-friendly revision materials a JSS1 learner could receive.
- `C0` The pack is well-structured, uses student-friendly language, and provides enough scaffolding for a JSS1 learner to work through it independently at home; the only weakness is that the inclusion of factorisation and equation-solving may confuse students who have not been taught those topics yet.

- **C1** The pack is exceptionally well structured for independent home use: language is simple, worked examples are thorough, practice items progress in difficulty, and full annotated answers allow self-correction — all highly appropriate for JSS1 learners.
- **C1** The pack is exceptionally well-structured for independent home use: clear rules, model examples, practice questions, and fully explained answers allow a JSS1 student to revise each topic and check their own understanding without any additional support.
- **C1** The pack is exceptionally well-structured for independent home use: clear rules, abundant worked steps, matched practice questions, and a diagnostic answer key combine to make it highly useful for JSS1 learners revising without teacher support.
- **C1** The pack is exceptionally well-suited for independent home use: language is accessible, examples are stepped and annotated, practice questions progress logically, and complete answer keys enable self-assessment. A JSS1 learner working alone should be able to follow it without additional support.
- **C1** The pack is exceptionally well structured for independent use: language is simple and age-appropriate, the progression from explanation to example to practice is logical, and the answer key with reasoning allows students to identify and correct their own mistakes.
- **C1** The pack is exceptionally well-structured for unsupported home use: clear explanations, progressive worked examples, and complete answer keys with reasoning enable a JSS1 student to review, practise, and self-correct all three topics independently.
- **C2** The pack is exceptionally well-structured for independent home use: clear rules, stepped examples with explanations, varied practice questions, and a complete answer key with working all combine to make it genuinely useful for a JSS1 learner revising alone.
- **C2** The pack is well-organised, uses plain language appropriate for JSS1, and provides enough scaffolding (clear rules, worked examples at increasing difficulty, and fully explained answers) for a student to revise the entire lesson independently without a teacher present.
- **C2** The pack is exceptionally well-structured for unsupported home use: clear language, logical progression, multiple worked examples, and a fully explained answer key make it highly useful for JSS1 self-revision.
- **C2** The pack is well-structured for independent use: clear rules, multiple graded examples, matched practice questions, and complete answer explanations together give a JSS1 student everything needed to review and self-check without teacher support.
- **C2** The pack is exceptionally well-structured for independent home use: clear rules, graded examples, practice questions drawn straight from the lesson, and detailed answer keys all combine to make it highly accessible and useful for a JSS1 student revising alone.
- **C2** The pack is exceptionally well-organised for independent home use: clear language, progressive examples, and detailed answer keys with reasoning make it highly useful for JSS1 students revising without teacher support.
- **C3** The pack is exceptionally well-structured for independent home use: clear rules, staged examples, matched practice, and complete answers make it highly accessible and useful for a JSS1 learner revising alone.
- **C3** The pack is well-structured, self-contained, and written in accessible language with full solutions — making it genuinely useful as standalone take-home revision material for JSS1 students.
- **C3** The pack is exceptionally well structured for independent home use: clear rules, graduated examples, matched practice, and a complete answer key with explanations make it highly accessible and useful for a JSS1 learner.
- **C3** The pack is well-structured, uses simple language appropriate for JSS1, progresses from explanation through examples to practice with answers, and is entirely self-contained for independent revision at home.
- **C3** The pack is exceptionally well-structured for independent home use: rules are stated plainly, examples are fully worked with explanations, and detailed answers allow students to diagnose and correct their own errors.
- **C3** The pack is well-structured for independent home use: language is accessible to JSS1 pupils, worked examples build in difficulty, and detailed answer keys with reasoning make it genuinely self-sufficient revision material.